

Agentic LLMs and AI for Theorem Proving

Computer-Assisted Theorem Proving (7CCSACTP)

Dr. Mohammad Abdulaziz

Department of Informatics, King's College London

Week 5

1. History: AI & Theorem Proving

▶ AI for Theorem Proving:

- ▶ AI has been considered as a tool to improve theorem proving for a long time by now.

▶ First & Most Impactful Attempt:

- ▶ **"Sledgehammer"** in Isabelle/HOL.
- ▶ First developed by Larry Paulson and Jasmin Blanchette [Paulson and Blanchette, 2010].
- ▶ Combines external Automated Theorem Provers (ATPs) and SMT solvers to automate subgoals.

▶ Tutorial & Practical Usage:

- ▶ Sledgehammer invocation and automated proof reconstruction in Isar.

2026-08-12

Agentic AI for Theorem Proving

└ 1. History: AI & Theorem Proving

- AI has been considered as a tool to improve theorem proving for a long time by now.
- The first and most impactful attempt was the development of Sledgehammer for Isabelle by Larry Paulson et al. [Paulson and Blanchette, 2010].
- Sledgehammer bridges interactive theorem proving with automated theorem provers (ATPs) and SMT solvers.

1. History: AI & Theorem Proving

- ▶ **AI for Theorem Proving:**
 - ▶ AI has been considered as a tool to improve theorem proving for a long time by now.
- ▶ **First & Most Impactful Attempt:**
 - ▶ **"Sledgehammer"** in Isabelle/HOL.
 - ▶ First developed by Larry Paulson and Jasmin Blanchette [Paulson and Blanchette, 2010].
 - ▶ Combines external Automated Theorem Provers (ATPs) and SMT solvers to automate subgoals.
- ▶ **Tutorial & Practical Usage:**
 - ▶ Sledgehammer invocation and automated proof reconstruction in Isar.

- ▶ **Custom Fine-Tuning & Reinforcement Learning (RL):**
 - ▶ Systems train custom models using compiler pass/fail signals as RL rewards.
 - ▶ Key players: **DeepMind** (AlphaProof / Gemini DeepThink), **Harmonic** (Aristotle), and **Axiom Math** (AxiomProver / AXLE).
- ▶ **Olympiad & Benchmark Breakthroughs:**
 - ▶ Benchmarks: **PutnamBench** and **IMO** were mathematical holy grails.
 - ▶ **IMO Gold** achieved by AI in 2025 (by DeepMind and Harmonic).
 - ▶ **PutnamBench 2025** currently solved by ≥ 20 proving systems.

2. Custom Models for Formal Proving

- ▶ **Custom Fine-Tuning & Reinforcement Learning (RL):**
 - ▶ Systems train custom models using compiler pass/fail signals as RL rewards.
 - ▶ Key players: **DeepMind** (AlphaProof / Gemini DeepThink), **Harmonic** (Aristotle), and **Axiom Math** (AxiomProver / AXLE).
- ▶ **Olympiad & Benchmark Breakthroughs:**
 - ▶ Benchmarks: **PutnamBench** and **IMO** were mathematical holy grails.
 - ▶ **IMO Gold** achieved by AI in 2025 (by DeepMind and Harmonic).
 - ▶ **PutnamBench 2025** currently solved by ≥ 20 proving systems.

2. Custom Models for Formal Proving

- Proprietary systems initially dominated formal theorem proving by training custom models.
- Benchmarks like PutnamBench and the International Mathematical Olympiad (IMO) were considered mathematical holy grails.
- DeepMind developed AlphaProof by fine-tuning Gemini models with reinforcement learning (RL) search in Lean.
- Specialised startups like Harmonic (Aristotle) and Axiom Math (AxiomProver / AXLE) trained custom transformer policy and value models using RL feedback loops with theorem prover compilers.
- IMO Gold was achieved by AI in 2025 by DeepMind and Harmonic.
- PutnamBench 2025 is currently solved by at least 20 different proving systems.

- ▶ **Inaccessibility of Proprietary Custom Models:**
 - ▶ Heavy RL infrastructure requires multi-million-dollar compute clusters.
- ▶ **Standard Coding Agents + Theorem Prover Loops:**
 - ▶ Josef Urban showed standard coding agents (e.g. Claude Code, Open Code, Agy) can formalise entire textbooks (e.g. Munkres Topology) for $\approx$ \$200/month.
- ▶ **Agentic Interactive Workflow:**
 - ▶ Connects off-the-shelf LLMs directly to provers via Model Context Protocol (MCP).
 - ▶ Eliminates the need for pre-training or fine-tuning custom models.

3. Democratisation via Agentic Proving

- Proprietary custom models were extremely expensive and inaccessible to students or individual researchers.
- Josef Urban and collaborators discovered that standard, off-the-shelf LLM coding agents are remarkably capable when connected to interactive theorem provers.
- In a recent landmark experiment, standard coding agents formalised an entire 800+ theorem textbook on Topology (Munkres) for $\approx$ \$200/month.
- This connects off-the-shelf agents via Model Context Protocol (MCP) servers, eliminating the multi-million-dollar cost of training custom foundation models.

3. Democratisation via Agentic Proving

- ▶ **Inaccessibility of Proprietary Custom Models:**
 - ▶ Heavy RL infrastructure requires multi-million-dollar compute clusters.
- ▶ **Standard Coding Agents + Theorem Prover Loops:**
 - ▶ Josef Urban showed standard coding agents (e.g. *Claude Code*, *Open Code*, *Agy*) can formalise entire textbooks (e.g. *Munkres Topology*) for $\approx$ \$200/month.
- ▶ **Agentic Interactive Workflow:**
 - ▶ Connects off-the-shelf LLMs directly to provers via Model Context Protocol (MCP).
 - ▶ Eliminates the need for pre-training or fine-tuning custom models.

4. Current State & Industrial Adoption

4. Current State & Industrial Adoption

- ▶ **Commercial Ecosystem & Venture Funding:**
 - ▶ Specialised companies: Harmonic, Math Inc, Axiom Math, etc. (securing major VC funding).
- ▶ **Industrial Formal Verification:**
 - ▶ Adoption for mission-critical software & hardware verification (e.g. AWS, Apple, Arm, Intel).
- ▶ **Integration in Mathematical Research:**
 - ▶ Top mathematicians (e.g. Terence Tao) incorporating agentic formal provers into active research.

- The agentic proving revolution has rapidly transitioned into commercial and industrial application.
- Specialised companies have secured significant venture capital funding (e.g. Harmonic, Math Inc, Axiom Math).
- Major technology companies have adopted formal verification for mission-critical software and hardware systems (e.g. AWS, Apple, Arm, Intel).
- Top mathematicians like Terence Tao have actively incorporated agentic formal provers into mainstream mathematical research.

4. Current State & Industrial Adoption

- ▶ **Commercial Ecosystem & Venture Funding:**
 - ▶ Specialised companies: *Harmonic*, *Math Inc*, *Axiom Math*, etc. (securing major VC funding).
- ▶ **Industrial Formal Verification:**
 - ▶ Adoption for mission-critical software & hardware verification (e.g. AWS, Apple, Arm, Intel).
- ▶ **Integration in Mathematical Research:**
 - ▶ Top mathematicians (e.g. Terence Tao) incorporating agentic formal provers into active research.

5. Technical Setup: Agentic LLMs for Theorem Proving

► Integrating Theorem Provers with LLMs:

- Connect a coding agent (e.g. *Claude Code*, *Open Code*, *Codex*, *Agy*) directly to Isabelle.

► Model Context Protocol (MCP) Architecture:



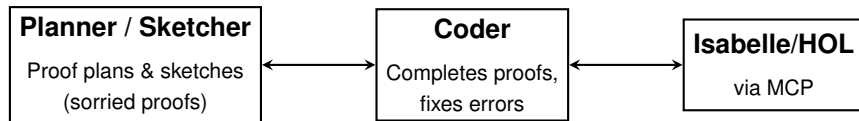
► MCP Implementations & Repositories:

- Programmatic actions: `write-file`, `process-file`, `save-file`, etc.
- **PIDE-MCP**: github.com/kappelmann/isabelle-pide-mcp
- **Isabelle/Q**: github.com/aws-labs/AutoCorrode (AWS Labs)

- In this week I will demonstrate a new but strongly emerging way to apply theorem provers like Isabelle, via integration with LLMs.
- You should have a coding agent ready, e.g. Claude Code, Open Code, Codex, Agy, etc.
- In my demos I will use Claude Code.
- Main idea: Connect a coding agent to the theorem prover via a Model Context Protocol (MCP) server.
- An MCP server exposes Isabelle programmatically: `write-file`, `process-file`, `save-file`, etc.
- We will use PIDE-MCP (<https://github.com/kappelmann/isabelle-pide-mcp>) and AWS's Isabelle/Q (<https://github.com/aws-labs/AutoCorrode>).

6. Agentic Proving Architectures

Basic Two-Stage Agentic Architecture [Jiang et al., 2023]:



► Workflow & Feedback Loop (Draft, Sketch & Prove):

- Planner creates structured proof sketches (`sorry`) [Jiang et al., 2023].
- Coder fills in Isar proof steps & verifies against Isabelle via MCP.
- Unresolved errors loop back from Coder to Planner.

► Further Reading & LGT Discussion:

- As the field is moving very fast, a paper will be provided on **KEATS** to read and discuss in the **LGT**.

6. Agentic Proving Architectures

- The most basic agentic architecture for theorem proving consists of a Planner/Sketcher, a Coder, an MCP server, and Isabelle.
- Introduced in the "Draft, Sketch, and Prove" paradigm by Jiang, Li et al. [Jiang et al., 2023].
- The Planner/Sketcher produces detailed proof plans and sketches (e.g. sorried proofs).
- The Coder takes the proof plans & sketches, completes them, and ensures Isabelle accepts them via MCP.
- If an error cannot be resolved by the Coder, it passes back to the Planner/Sketcher.
- Since the area is moving very fast, I will leave a paper on KEATS for you to read and discuss in the LGT.

7. Closing Remarks & Coursework Guidelines

► Core Skills You Must Master:

1. Splitting problems into subproblems.
2. Formally modelling problem domains.
3. Formulating correct specifications.
4. Constructing correct, concise, fast-checking proofs (Isabelle linter).

► LLM Usage Policy for Coursework:

- **Design & Sketches:** Allowed for exploring design alternatives (must justify choices).
- **Lemma Proving:** Allowed *only* for facts referencing standard library definitions. **Not allowed** for facts about your own defined constants.

► Assessment (50% Coursework + 50% Exam):

- **Coursework (50%):** Includes a **Viva** to explain design choices.
- **Exam (50%):** Paper-based written examination.

2026-08-12

Agentic AI for Theorem Proving

7. Closing Remarks & Coursework Guidelines

- You still need to understand technical details yourself: (1) splitting problems, (2) formal modelling, (3) specifications, (4) proving properties (correct, readable/concise, fast checking).
- Module Guidelines for LLM use:
 1. Design decisions & proof sketches: Explore alternatives and justify choices.
 2. Proving lower-level facts: Allowed **ONLY** for facts referencing standard Isabelle distribution definitions. Do **NOT** use LLMs for facts about your own defined constants.
- Coursework includes a Viva where you must defend your design choices.
- Assessment: 50

7. Closing Remarks & Coursework Guidelines

- **Core Skills You Must Master:**
 1. Splitting problems into subproblems.
 2. Formally modelling problem domains.
 3. Formulating correct specifications.
 4. Constructing correct, concise, fast-checking proofs (Isabelle linter).
- **LLM Usage Policy for Coursework:**
 - **Design & Sketches:** Allowed for exploring design alternatives (must justify choices).
 - **Lemma Proving:** Allowed *only* for facts referencing standard library definitions. **Not allowed** for facts about your own defined constants.
- **Assessment (50% Coursework + 50% Exam):**
 - **Coursework (50%):** Includes a Viva to explain design choices.
 - **Exam (50%):** Paper-based written examination.

References I

Albert Q. Jiang, Sean Welleck, Jin Peng Zhou, Wenda Li, Jiacheng Liu, Mateja Jamnik, Timothée Lacroix, Yuhuai Wu, and Guillaume Lample. Draft, Sketch, and Prove: Guiding Formal Theorem Provers with Informal Proofs, 2023.

Lawrence C Paulson and Jasmin Christian Blanchette. Three years of experience with sledgehammer, a practical link between automatic and interactive theorem provers. In *PAAR IJCAR*, 2010.

2026-08-12

References

References I

Albert Q. Jiang, Sean Welleck, Jin Peng Zhou, Wenda Li, Jiacheng Liu, Mateja Jamnik, Timothée Lacroix, Yuhuai Wu, and Guillaume Lample. Draft, Sketch, and Prove: Guiding Formal Theorem Provers with Informal Proofs, 2023.

Lawrence C Paulson and Jasmin Christian Blanchette. Three years of experience with sledgehammer, a practical link between automatic and interactive theorem provers. In *PAAR IJCAR*, 2010.